ML2 2.3

Complex Machine Learning Model

Random Forest

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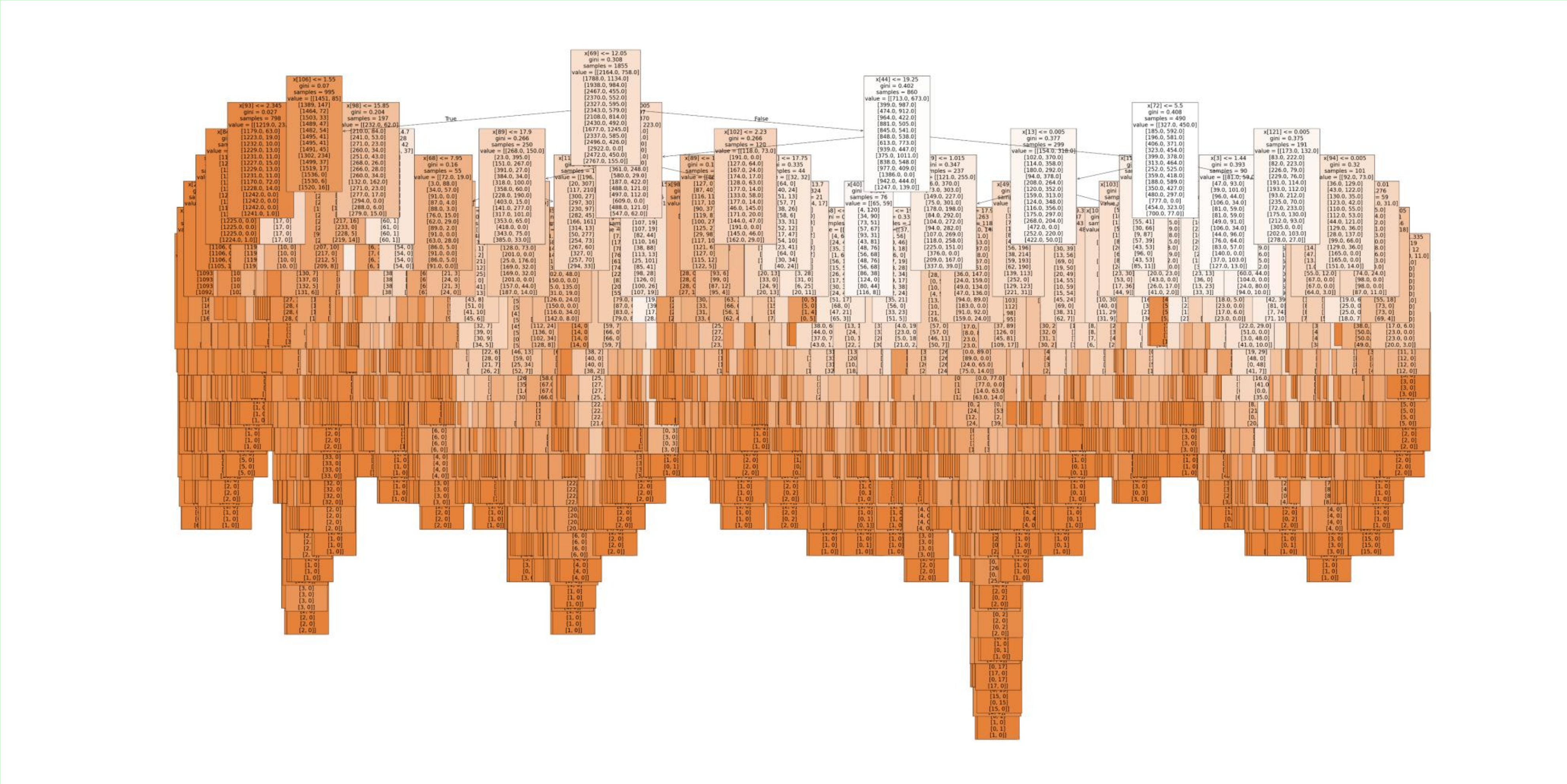
Random Forest

- In the first Random Forest I ran, I selected a single decade to examine.
- I chose decade 1960
- This Random Forest had all 15 weather stations and all 15 pleasant weather predictions.
- This RF is quite large and obviously difficult to read.
- Our Model Accuracy was: 0.6032831737346102



Random Forest Feature Importance

- When creating Feature Importance there were 135 values.
- Looking within a range of 0.042943 – 0.011188 (top 29 items) we can see that Temp_Max, Temp_Mean and Precipitation are the top features with the most influence.

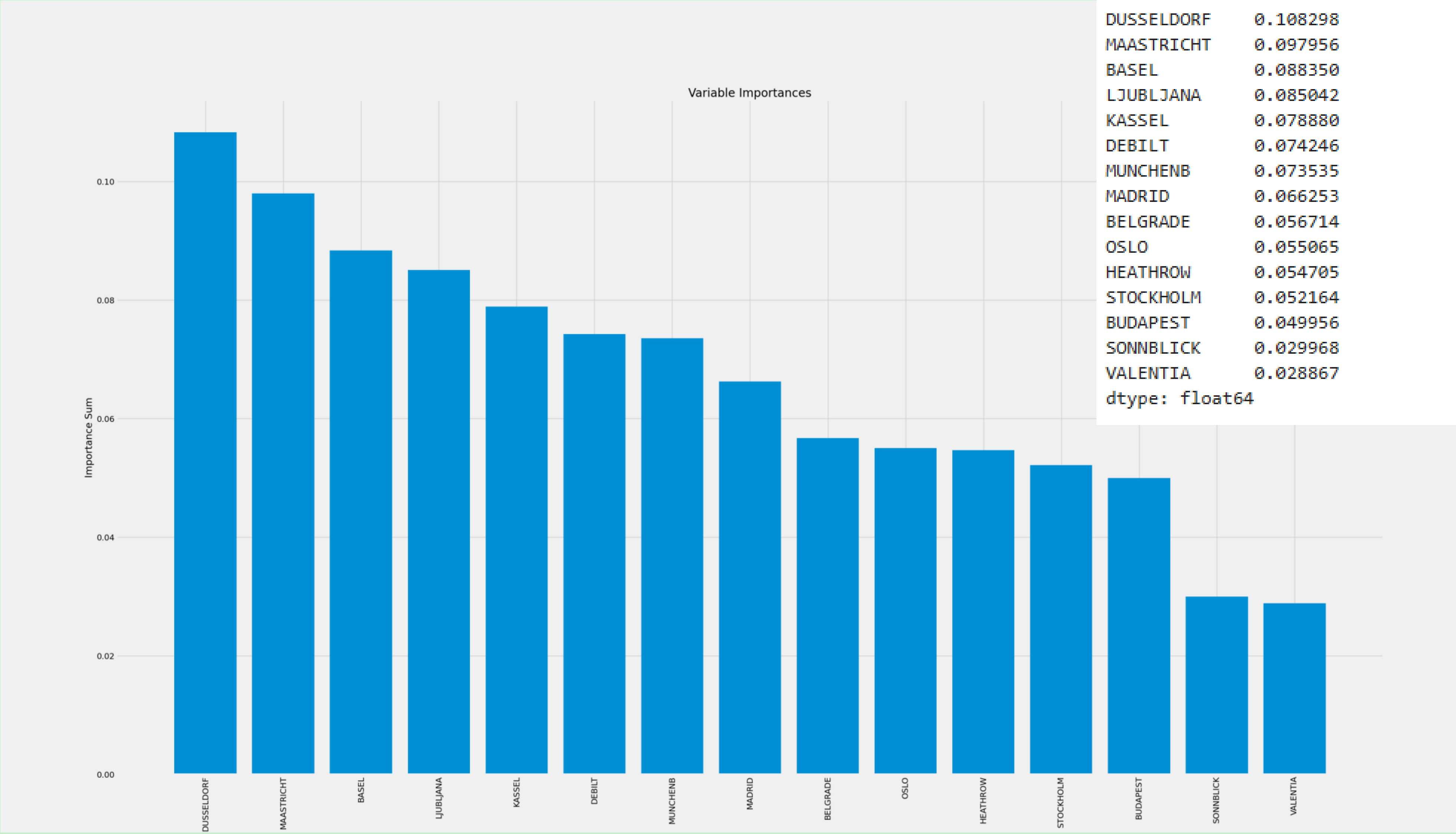
```
# Looking at all the important features in

important.sort_values(ascending=False)
```

MAASTRICHT_temp_max	0.042943
DUSSELDORF_temp_max	0.037980
BASEL_temp_max	0.035093
DUSSELDORF_temp_mean	0.032047
KASSEL_temp_max	0.024634
LJUBLJANA_temp_mean	0.024563
DEBILT_temp_max	0.024301
LJUBLJANA_temp_max	0.020997
MUNCHENB_temp_max	0.020697
KASSEL_precipitation	0.018618
DUSSELDORF_precipitation	0.018457
HEATHROW_temp_mean	0.017576
LJUBLJANA_precipitation	0.017541
BUDAPEST_temp_max	0.017312
MAASTRICHT_precipitation	0.017294
BELGRADE_precipitation	0.017202
BASEL_precipitation	0.016351
BASEL_temp_mean	0.015653
MADRID_temp_max	0.015583
DEBILT_precipitation	0.014862
MUNCHENB_precipitation	0.014309
MADRID_temp_mean	0.014286
STOCKHOLM_temp_max	0.013997
BUDAPEST_precipitation	0.013753
OSLO_temp_max	0.013007
MAASTRICHT_temp_mean	0.012978
KASSEL_temp_mean	0.012170
DEBILT_temp_mean	0.011806
MUNCHENB_temp_mean	0.011188
BELGRADE_temp_mean	0.009138
BELGRADE_temp_max	0.009063
HEATHROW_temp_max	0.009005
OSLO_temp_min	0.008483
STOCKHOLM_temp_mean	0.008300
OSLO_temp_mean	0.007843
MADRID_precipitation	0.007839
HEATHROW_precipitation	0.007551
BUDAPEST_temp_mean	0.006886
OSLO_global_radiation	0.006691
MADRID_temp_min	0.006487
STOCKHOLM_global_radiation	0.006411

Random Forest Bar Chart

- Our bar chart shows the weather stations with the most influence (left to right)
- We can even look at the sum of each weather station to visualize it’s rank.

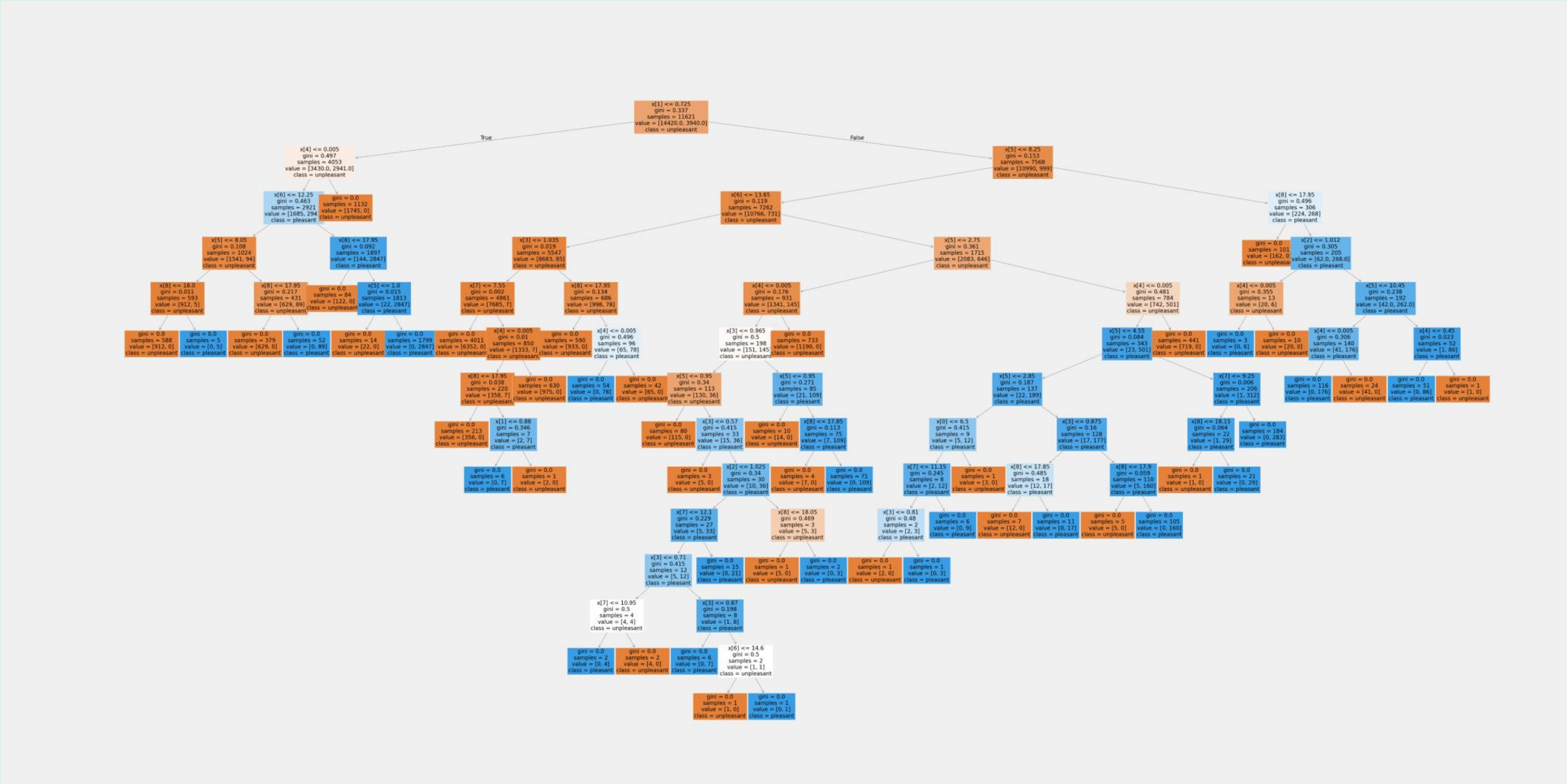


DUSSELDORF	0.108298
MAASTRICHT	0.097956
BASEL	0.088350
LJUBLJANA	0.085042
KASSEL	0.078880
DEBILT	0.074246
MUNCHENB	0.073535
MADRID	0.066253
BELGRADE	0.056714
OSLO	0.055065
HEATHROW	0.054705
STOCKHOLM	0.052164
BUDAPEST	0.049956
SONNBLICK	0.029968
VALENTIA	0.028867

dtype: float64

Dusseldorf

- I began running RF for the top 3 weather stations. Dusseldorf was the highest influencing weather station according to the first RF.
- This RF has all the years within the data but only the observations for Dusseldorf
- This RF is quite a bit smaller and a lot easier to read.
- Our Model Accuracy was: 1.0 – I assume simply because we are working with one weather station.



Dusseldorf
Feature Importance

- When creating Feature Importance there were 9 values.
- We can see that like in the main RF, our top influencing features are precipitation, temperature max and temperature mean.
- Our sum for Dusseldorf is 0.108

```
# Creating a list of features importance

important = pd.Series(clf.feature_importances_, index = dusseldorf_features)
important.sort_values(ascending=False)
```

```
DUSSELDORF_precipitation      0.348972
DUSSELDORF_temp_max           0.231966
DUSSELDORF_temp_mean          0.131520
DUSSELDORF_global_radiation   0.104448
DUSSELDORF_sunshine           0.089203
DUSSELDORF_cloud_cover        0.035935
DUSSELDORF_temp_min           0.030351
DUSSELDORF_humidity           0.016428
DUSSELDORF_pressure           0.011177
dtype: float64
```

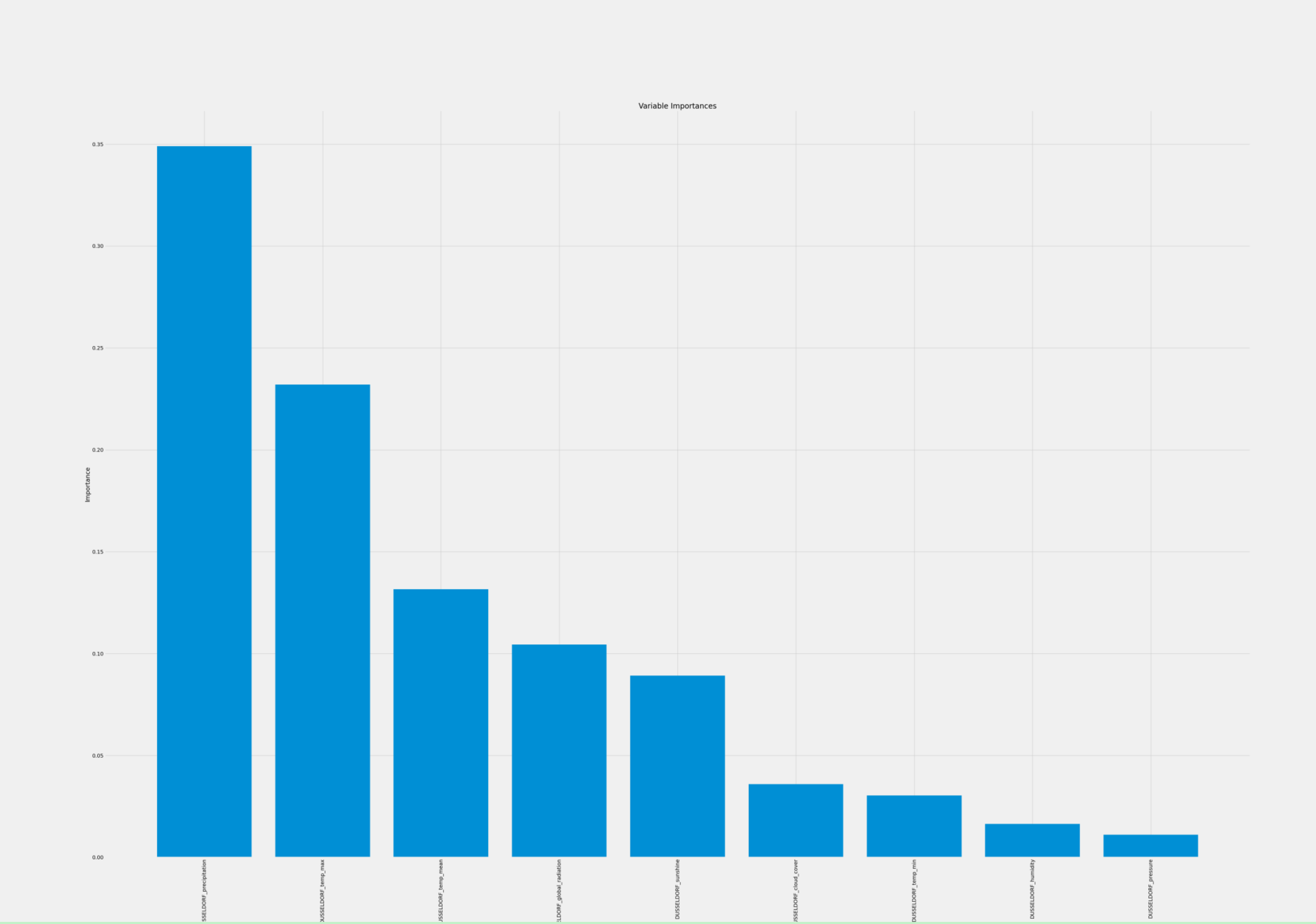
```
dusseldorf_sum = {
    'DUSSELDORF': DUSSELDORF.sum()}
```

```
dusseldorf_sum
```

```
{'DUSSELDORF': 0.10829766769077326}
```

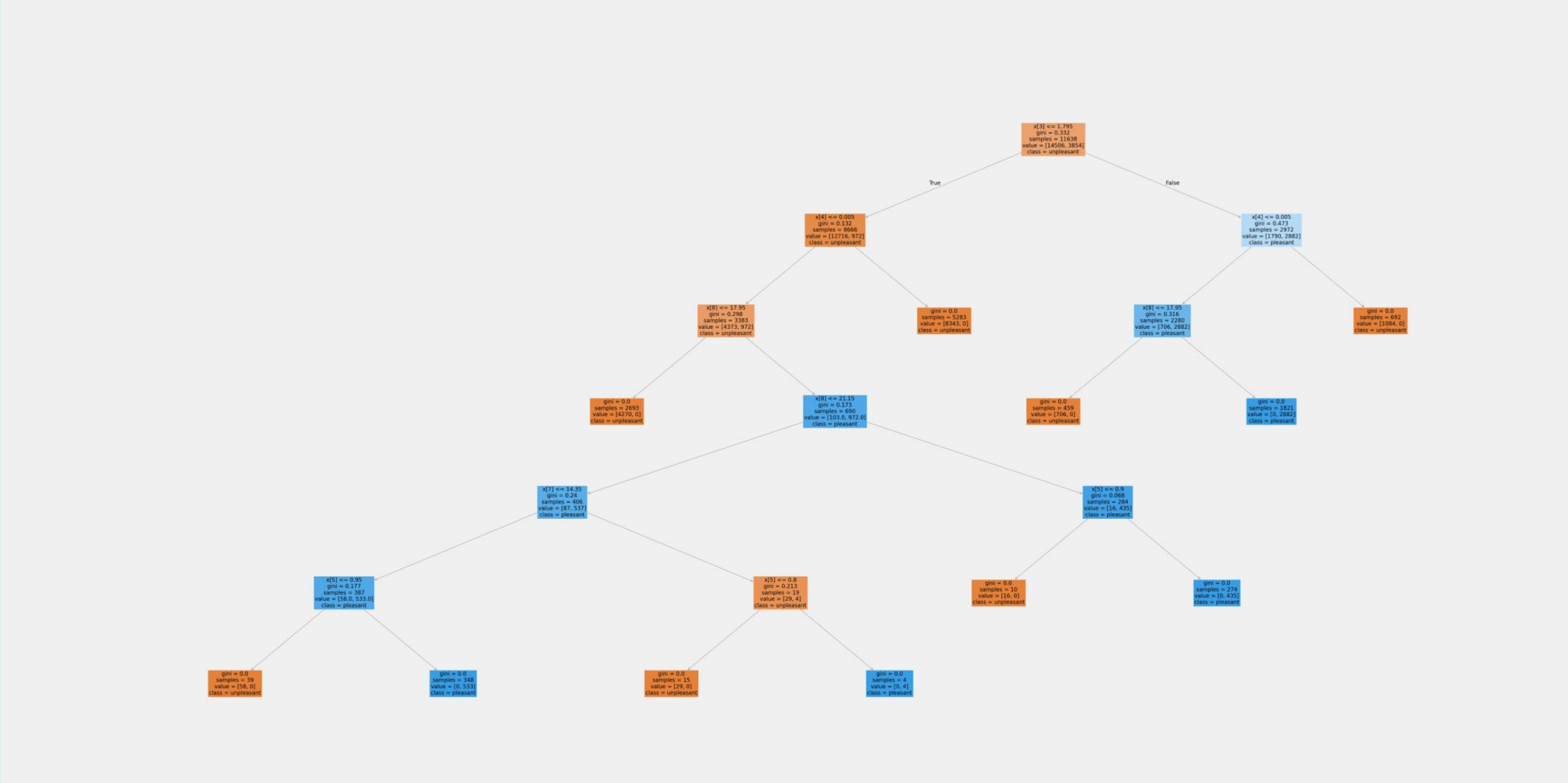
Dusseldorf Bar Chart

- Our bar graph matches nicely with our feature importance.
- Precipitation has a large influence with Temperature max coming in as strong second.
- Temperature mean is less than half of precipitation so there's not a large amount of influence.



Maastricht

- Running the second weather station with influence and this RF is so simple.
- Again, this RF has all the years available but, we’re only focusing on Maastricht weather station.
- Our Model Accuracy was: 1.0 – I assume simply because we are working with one weather station.



Maastricht
Feature Importance

- When creating Feature Importance there were 9 values.
- Again, we can see that like in the main RF, our top influencing features are precipitation, temperature max and temperature mean.
- Our sum for Maastricht is 0.0979

```
# Creating a list of features importance

important = pd.Series(clf.feature_importances_, index = maastricht_features)
important.sort_values(ascending=False)
```

```
MAASTRICHT_precipitation       0.318355
MAASTRICHT_temp_max            0.297020
MAASTRICHT_temp_mean           0.130205
MAASTRICHT_global_radiation    0.091165
MAASTRICHT_sunshine            0.074430
MAASTRICHT_cloud_cover         0.037596
MAASTRICHT_humidity            0.022949
MAASTRICHT_temp_min            0.017309
MAASTRICHT_pressure            0.010972
dtype: float64
```

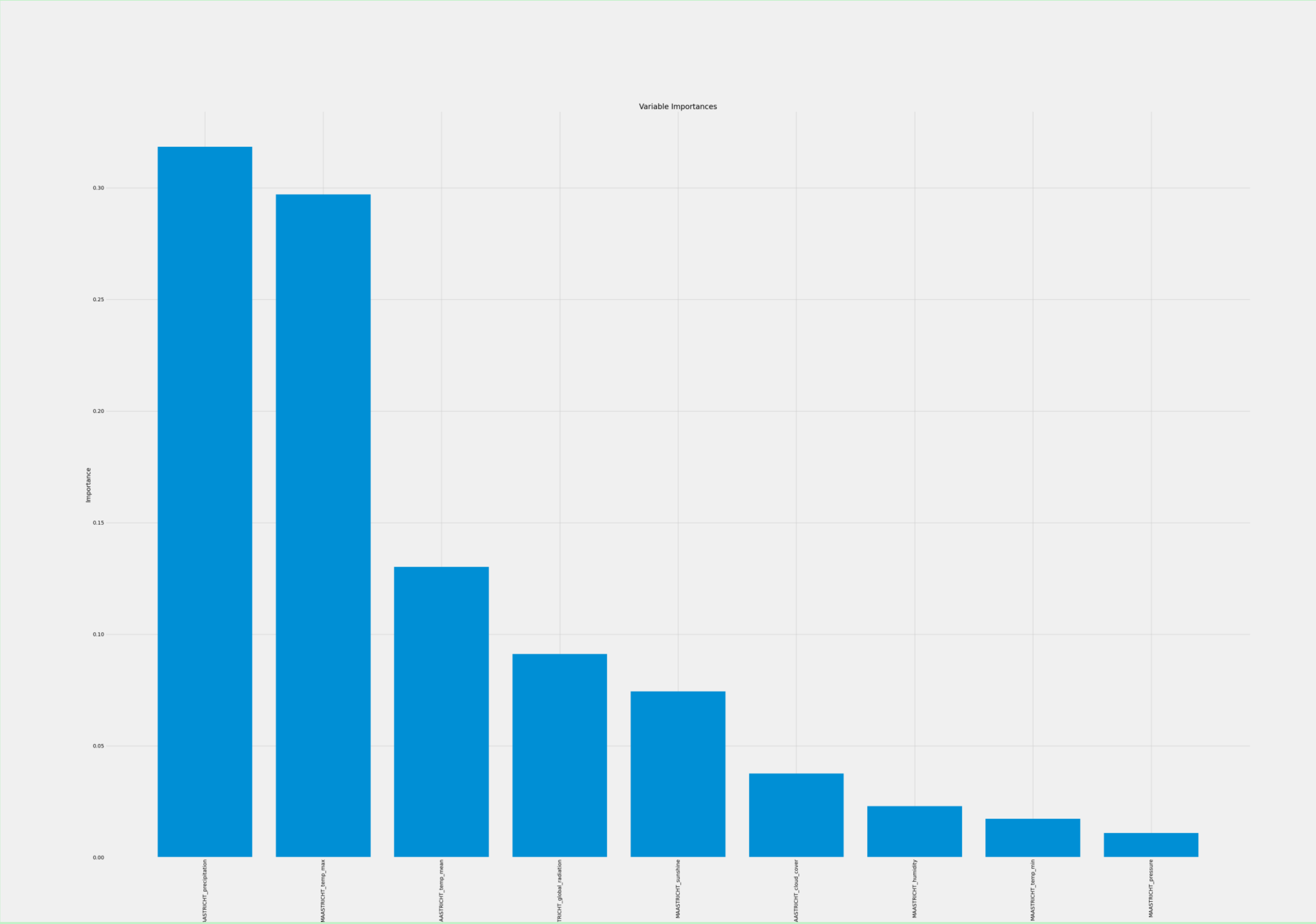
```
maastricht_sum = {
    'MAASTRICHT': MAASTRICHT.sum()}
```

```
maastricht_sum
```

```
{'MAASTRICHT': 0.09795604435275818}
```

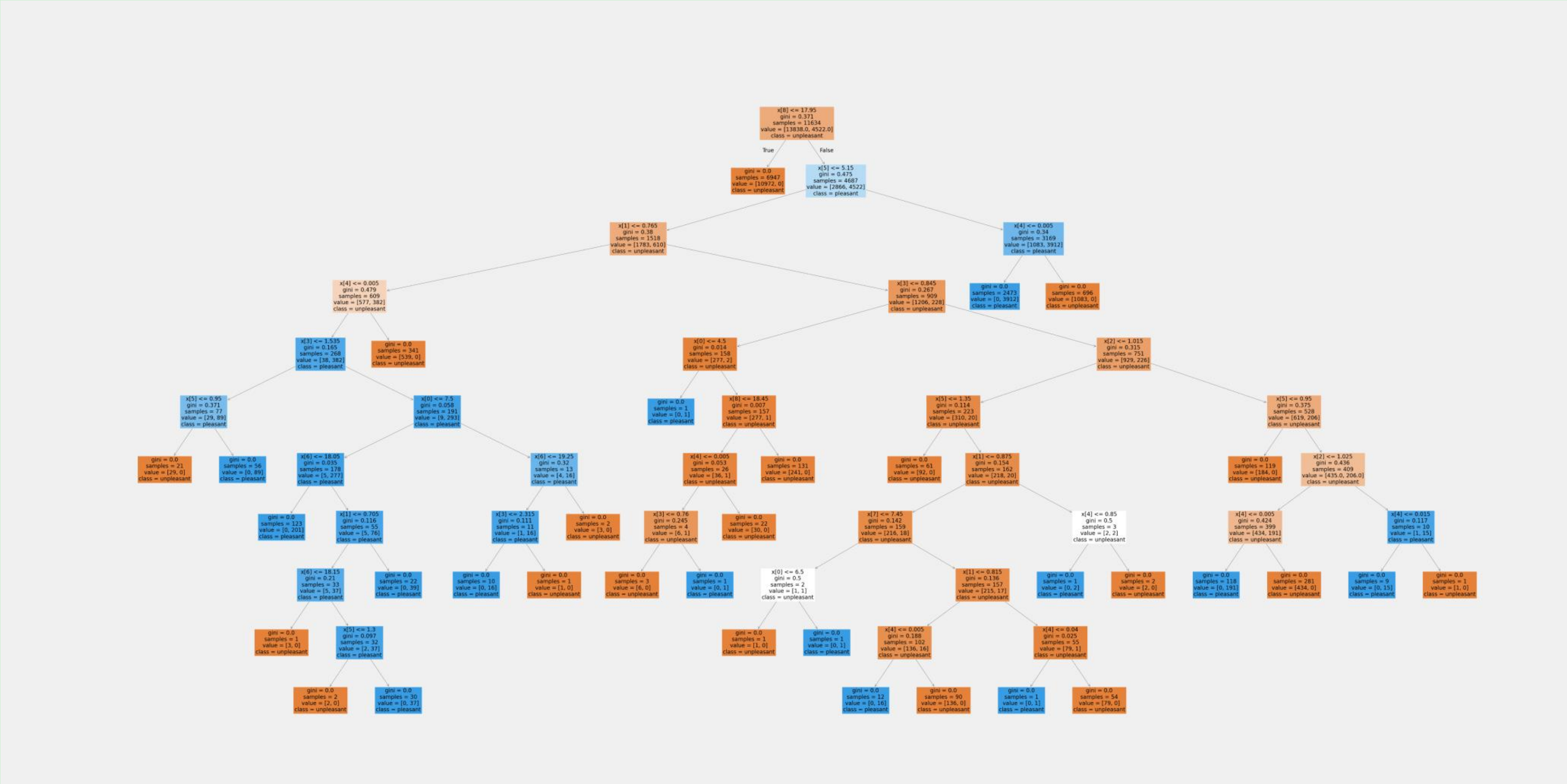
Maastricht Bar Chart

- Our bar graph matches nicely with our feature importance.
- Precipitation has a large influence with Temperature max as an extremely strong runner up!
- Temperature mean is less than half of precipitation so there’s not a large amount of influence, either.
- Comparing Dusseldorf with Maastricht we are starting to see that temperature mean isn’t as influential as we thought.



Basel

- Our third weather station and this RF is a little busier than the previous two weather stations but, it’s not too bad..
- This RF has all the years within the data but only the observations for Basel
- Our Model Accuracy was: 1.0 – I assume simply because we are working with one weather station.



Basel
Feature Importance

- We are still observing 9 Feature Importance .
- We can see that like in the main RF, our top influencing features are precipitation, temperature max but Basel offers a difference!
- Basel third feature is global radiation. This would lead me to believe that perhaps Basel is closer to the equator.
- Our sum for Basel is 0.088

```
# Creating a list of features importance

important = pd.Series(clf.feature_importances_, index = basel_features)
important.sort_values(ascending=False)
```

```
BASEL_precipitation       0.350921
BASEL_temp_max            0.267804
BASEL_global_radiation    0.116226
BASEL_sunshine            0.104025
BASEL_temp_mean           0.078280
BASEL_temp_min            0.039065
BASEL_cloud_cover         0.024233
BASEL_humidity            0.011451
BASEL_pressure            0.007996
dtype: float64
```

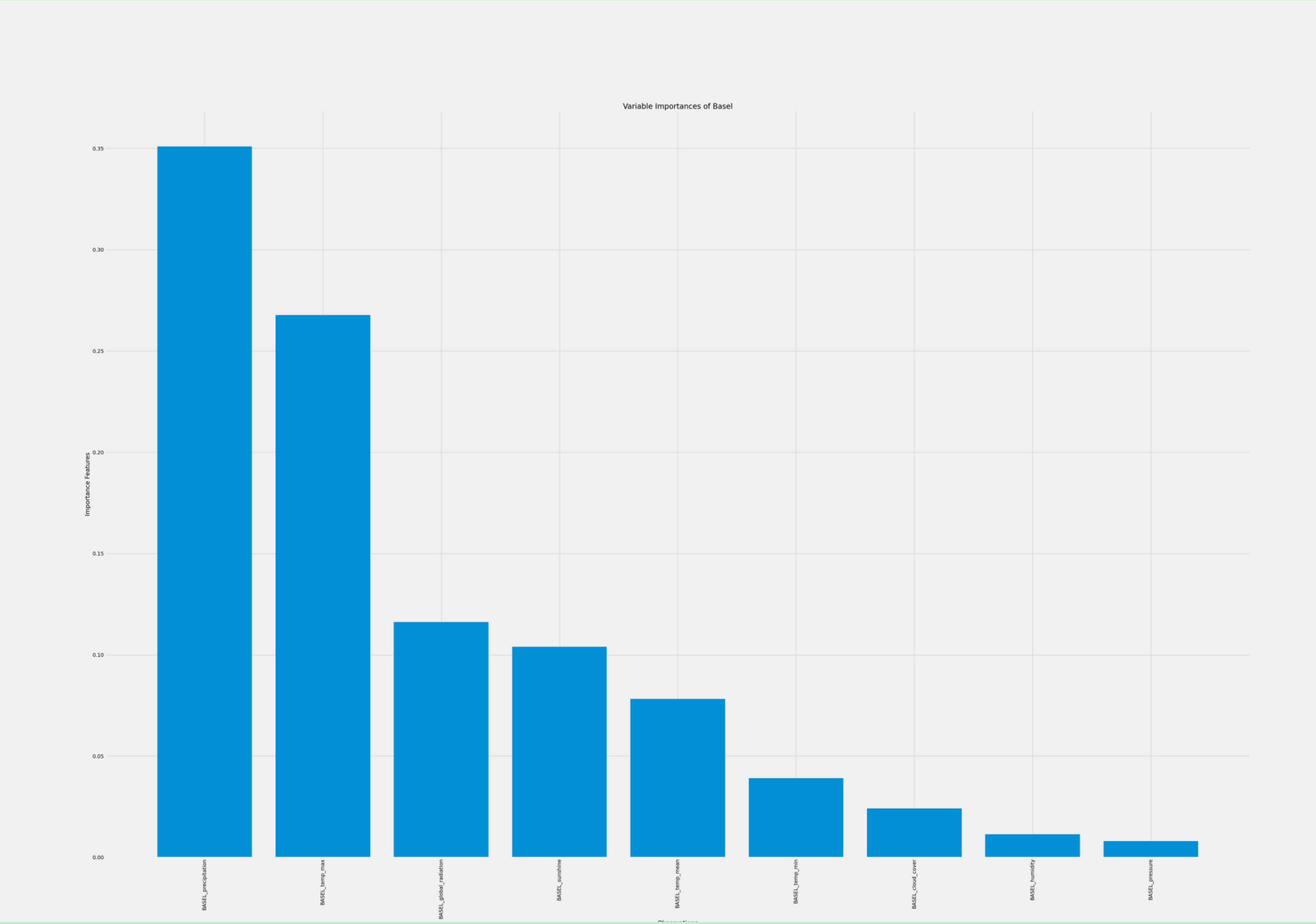
```
basel_sum = {
    'BASEL': BASEL.sum()}
```

```
basel_sum
```

```
{'BASEL': 0.0883496021719356}
```

Basel
Bar Chart

- Our bar graph matches nicely with our feature importance.
- Precipitation has a large influence with Temperature max coming in as strong second.
- The Global Radiation is less than half of precipitation so there’s not a large amount of influence.



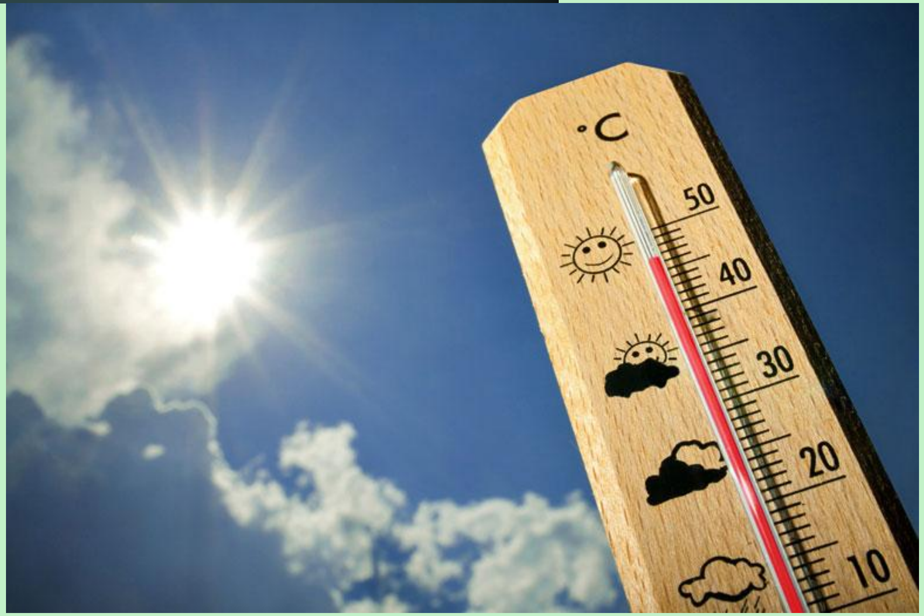
Conclusion

Our random Forest and Feature Importance have begun to point us in the right direction for understanding what influences a good day.

With precipitation, otherwise knowns as rain or hail, we can conclude that when the day is wet and rainy or even snows the day is not aligned with “pleasant” weather.

When a day has high temperatures, those days are also not considered “pleasant” weather.

Precipitation and temperature max have been our top influencers of unpleasant and pleasant weather. If these two parameters are low then we are more likely to have a day with pleasant weather.



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